

Database Management Systems (DBMS)

Examination Review Guide

Study Reference

Overview

Subject: Database Theory and SQL

Format: A4 Comprehensive Guide

Content: Multiple Choice, Case Study Analysis, and SQL Syntax

Part I: Multiple Choice Questions

1. DBMS provides interface between:
 - a. Data & files.
 - **b. Users & data.**
 - c. Applications programs & data.
 - d. None.
2. What happens when a DELETE statement is executed on a table without a WHERE clause?
 - a. An error is generated and no records are deleted.
 - **b. All records in the table are deleted.**
 - c. The table structure is deleted, but the data is retained.
 - d. The first record in the table is deleted.
3. Refer to below statements about primary key in a table. Find out which statement/s are true.
 - A) *Primary Key is used to identify each record uniquely in a table.*
 - B) *Each table can contain more than one primary key.*
 - **a. A is true and B is false.**
 - b. A is false and B is true.
 - c. A is true and B is true.
 - d. A is false and B is false.
4. Which one is the function of DBMS?
 - a. Data storage, retrieval and update.
 - b. Data integrity management.
 - **c. Both (a) and (b).**
 - d. None of the above.
5. Which of the following is a CORRECT way of inserting a new row into Staff table?

- a. **INSERT INTO Staff VALUES ('SG44', 'Anne', 'Jones', 'Assistant', NULL, NULL, 8100, 'B003');**
 - b. INSERT IN Staff VALUES ('SG44', 'Anne', 'Jones', 'Assistant', NULL, NULL, 8100, 'B003')
 - c. INSERT INTO (staffNo, fName, lName, position, salary, branchNo) VALUES ('SG44', 'Anne', 'Jones', 'Assistant', 8100, 'B003')
 - d. INSERT TO Staff (staffNo, fName, lName, position, salary, branchNo) VALUES ('SG44', 'Anne', 'Jones', 'Assistant', 8100, 'B003');
6. Which of the following is NOT a common SQL datatype?
- a. **COMPLEX.**
 - b. VARCHAR.
 - c. INT.
 - d. FLOAT.
7. What is a database?
- a. A collection of spreadsheets.
 - b. A collection of files.
 - c. A collection of folders.
 - d. **A collection of tables.**
8. Which of the following clauses is used in an UPDATE statement to specify the rows to be updated?
- a. **WHERE.**
 - b. HAVING.
 - c. FROM.
 - d. SET.
9. What is a file-based approach to data management?
- a. Storing data in a spreadsheet.
 - b. **Storing data in a flat file format.**
 - c. Storing data in a cloud-based system.
 - d. Storing data in a database.
10. What is a one-to-many relationship in an ER diagram?
- a. Each entity in one entity type can be associated with multiple entities in another entity type.
 - b. Each entity in one entity type can be associated with one or more entities in another entity type.
 - c. **Each entity in one entity type can be associated with multiple entities in another entity type, but each entity in the other entity type can be associated with only one entity in the first entity type.**
 - d. Each entity in one entity type can be associated with only one entity in another entity type.
11. Select the CORRECT statement regarding Literals.

- a. Character values and date values are format-sensitive.
 - b. All numeric literals must be enclosed in quotes.
 - **c. All non-numeric literals must be enclosed in single quotes.**
 - d. The default date format is 'DD-YY-MON'.
12. Which of the following does not offer from a database?
- **a. Ability to unauthorized deletion of necessary data.**
 - b. Ability to insert new data.
 - c. Ability to update old data.
 - d. Ability to retrieve data.
13. Which of the following is NOT a step in building conceptual data model?
- **a. Determine index of relationship.**
 - b. Determine multiplicity constraints on relationships.
 - c. Identify entity types.
 - d. Determine candidate and primary key attributes.
14. What is a relational database?
- **a. A database that uses tables to store data.**
 - b. A database that stores data in a hierarchical structure.
 - c. A database that stores data in a network of files.
 - d. A database that stores data in a single file.
15. Which of the following is an example of a conceptual design decision?
- **a. Determining the relationships between entities.**
 - b. Determining the physical storage location of data.
 - c. Determining which indexes should be created.
 - d. Determining which data types should be used for each attribute.
16. What is the purpose of drawing an ER diagram?
- a. To show the user interface of a database.
 - **b. To show the data model of a database.**
 - c. To show the logical structure of a database.
 - d. To show the physical structure of a database.
17. Select the correct order of the main three phases of database design.
- **a. Conceptual Design, Logical Design, Physical Design.**
 - b. Conceptual Design, Physical Design, Relational Design.
 - c. Conceptual Design, Logical Design, Relational Design.
 - d. Physical Design, Conceptual Design, Logical Design.
18. Which of the following is NOT an alternative way to represent multiplicity constraint?
- **a. *.***
 - b. 1..1

- c. 0..*
- d. 0..1

19. A command that lets you change one or more fields in a record is:

- a. Insert.
- **b. Modify.**
- c. None of the given answers.
- d. Look-up.

20. What is the syntax for selecting all columns from a table in MySQL?

- a. SELECT FROM table_name;
- b. SELECT ALL FROM table_name;
- c. SELECT table_name;
- **d. SELECT * FROM table_name;**

21. Select the correct statement from the statements given below.

- a. Relational database is structured to recognize relationships among stored items.
- b. MySQL, Oracle, and SQL Server are relational database software.
- c. Relational databases are the most common types.
- **d. All the statements above are true.**

22. Select the incorrect statement regarding an entity.

- a. An Entity is a single unique object in the real world.
- b. Every entity is made up of 'attributes'.
- c. An entity can be a place, person, object, event, or concept.
- **d. An entity may have attributes.**

23. A column of a table that points to the primary key of another table is known as:

- a. Composite key.
- b. Surrogate key.
- c. Alternate key.
- **d. Foreign key.**

24. Which of the following is an advantage of the file-based approach?

- a. Ability to easily manage complex data relationships.
- b. Ability to handle large volumes of data.
- c. Ability to enforce data access controls.
- **d. None of the above.**

25. Select the CORRECT statement regarding SQL statement.

- **a. Reserved words are a fixed part of SQL and must be spelt exactly as required.**
- b. SQL statement only consists of reserved words.
- c. Most components are case sensitive.
- d. All SQL statements are terminated by a semi-colon (;).

26. What is the purpose of the GROUP BY clause in a MySQL query?
- a. To sort the rows in ascending order.
 - **b. To group rows based on a specified column or expression.**
 - c. To join two or more tables together.
 - d. To limit the number of rows returned.
27. Attributes that exist only when there is an association between entities are known as:
- a. Compound attributes
 - **b. Relationship attributes**
 - c. Composite attributes
 - d. Multi-valued attributes
28. Which of the following statements defines a database properly?
- a. A hardware that interacts with the end users.
 - b. A collection of records.
 - c. An unorganized collection of structured data.
 - **d. A shared repository of logically related data**
29. Which of these is NOT a limitation of file based approach?
- a. Each program maintains its own set of data
 - **b. Improved data sharing among programs**
 - c. Programs are written to satisfy fixed set of queries
 - d. Same data is held in several places
30. A Candidate key that consists of two or more attributes is called a:
- a. Alternate key
 - b. Surrogate key
 - c. Simple key
 - **d. Composite key**
31. Attributes that can be calculated using the value of a related attribute are:
- a. Compound attribute
 - **b. Derived attribute**
 - c. Relationship attribute
 - d. Multi-valued attribute.
32. Select the correct order of the three main phases of database design process.
- a. Logical, Conceptual, Physical
 - b. Conceptual, Physical, Logical
 - c. Physical, Conceptual, Logical
 - **d. Conceptual, Logical, Physical**
33. ...integrity ensures that no attribute of a primary key can be null.

- a. Referencing
- **b. Entity**
- c. Foreign Key
- d. Referential

34. Each attribute has its own predefined value scope, which is known as:

- **a. Attribute domain**
- b. Attribute scope
- c. Attribute occurrence
- d. Attribute degree

35. Which is the subset of SQL commands used to design Database structures?

- a. Data Manipulation Language (DML)
- b. Both of DDL and DML
- c. None of DDL and DML
- **d. Data Definition Language (DDL)**

36. Which of the following SQL clauses is used to Remove tuples from a database table?

- a. DROP
- b. CLEAR
- **c. DELETE**
- d. REMOVE

37. If a non-null foreign key does not match any primary key, it violates:

- a. Null Value Rule
- **b. Referential Integrity Rule**
- c. Foreign Key Rule
- d. Entity Integrity Rule

38. The DBMS acts as an interface between ...and ...of an enterprise system.

- **a. Database application and the database**
- b. The user and the software
- c. Application and SQL
- d. Data and the DBMS

Part II: Descriptive Analysis & SQL

Case Study Analysis (Movie Database)

- **Entities:** Director, Movie, Actor.
- **Director:** One director directs one or many movies.
- **Movie Identifier:** movieid.
- **Actor Identifier:** actorid.
- **Composite Attributes:** name and address (Actor entity).
- **Multi-valued Attributes:** Mobile and email.
- **Derived Attribute:** Age.

Case Study Analysis (Bank)

- **Entities:** Bank, Branch, Account, Account_Type, Customer.
- **Primary Keys:** Branch_number, bank_code, account_number, customer_number, account_code.
- **Multi-valued attribute:** Contact_Number.
- **Composite attribute:** Address.
- **Entity Strength:** **Bank** is a strong entity; **Branch** and **Customer** are weak entities.

Technical Definitions

- **Foreign keys** are NOT typically shown on **designing** (conceptual) data models.
- **Foreign key constraint** defines: 'There cannot be any mismatched foreign keys to the base table'.

Data Type Selection

Column	Data Type
CustomerID	INT
OrderDate	DATE
OrderID	INT
Name	VARCHAR

Table 1: Recommended Data Types

SQL Syntax Reference

Question: Write down the accurate SQL statement to list all customers whose names include letter 'a' in them?

Answer: `SELECT * FROM Customer WHERE Name LIKE '%a%';`